ResearchGate

See discussions, stats, and author profiles for this publication at: https://www.researchgate.net/publication/389014496

Adaptive Neural Network for CO2 Reduction: A Predictive Model Integrating MCMC and GAT Techniques for Oil Vessel Supply Chain

Chapter · February 2025

DOI: 10.1007/978-981-96-2625-0_41

CITATIONS
0

READS
13

3 authors, including:

Lapo Chirici
University of Pisa

13 PUBLICATIONS **15** CITATIONS

SEE PROFILE

All content following this page was uploaded by Lapo Chirici on 24 February 2025.

The user has requested enhancement of the downloaded file.

Adaptive Neural Network for CO_2 Reduction: A Predictive Model Integrating MCMC and GAT Techniques for Oil Vessel Supply Chain

Lapo Chirici[1](✉), Yi Wang[2], and Kesheng Wang[3]

[1] Department of Computer Science, Università Di Pisa, Pisa, Italy
lapochirici@gmail.com
[2] Business School, University of Bedfordshire, Luton, UK
yi.wang@beds.ac.uk
[3] Department of Mechanical and Industrial Engineering, NTNU, Trondheim, Norway
kesheng.wang@ntnu.no

Abstract. This paper proposes an innovative machine learning algorithm that integrates Graph Attention Networks (GAT) with Monte Carlo Markov Chain (MCMC) techniques to optimize the Work Breakdown Structure (WBS) in crude oil vessel production. This novel approach aims to reduce CO_2 emissions, minimize lead times, and enhance cost savings within the supply chain. A synthetic dataset representing 15,000 companies in the oil and gas sector was used to test the algorithm. The results demonstrate potential improvements in key metrics, paving the way for more sustainable and efficient supply chain operations.

Keywords: Graph Theory · MCMC · Neural Network · GAT · Carbon Footprint

1 Introduction

The crude oil industry, particularly in the production of vessels, faces significant pressures to balance environmental sustainability with operational efficiency. The International Energy Agency (IEA) reports that maritime activities contribute approximately 2.5% of global CO_2 emissions, with crude oil vessels representing a substantial portion of this impact. As environmental regulations become more stringent, the industry must adopt innovative strategies to minimize its carbon footprint while maintaining competitive advantage. Current trends in the sector indicate a growing emphasis on integrating advanced technologies into supply chain operations to achieve these goals. For instance, there has been a significant increase in the adoption of digital technologies and AI-driven solutions to enhance operational efficiency and reduce emissions. According to recent industry analyses, the global maritime industry's move towards greener practices is expected to drive substantial investments in sustainable technologies, potentially reaching $1 trillion by 2030. This shift is partly driven by international regulatory frameworks, such as the IMO 2020 sulphur cap and the European Union's Emissions Trading System (ETS), which are pushing the industry towards lower emissions. This paper introduces

© The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2025
Y. Wang et al. (Eds.): IWAMA 2024, LNEE 1364, pp. 330–336, 2025.
https://doi.org/10.1007/978-981-96-2625-0_41

a novel approach to optimizing Work Breakdown Structure (WBS) configurations in crude oil vessel production by integrating Graph Attention Networks (GAT) and Monte Carlo Markov Chain (MCMC) techniques. The proposed methodology aims to reduce CO2 emissions and enhance overall supply chain performance by dynamically prioritizing key processes and materials. The integration of GATs, which are adept at managing the complexity of graph-structured data, with MCMC methods, known for their robustness in probabilistic simulations, offers a comprehensive and innovative solution for addressing the environmental and operational challenges faced by the crude oil vessel production industry.

2 Machine Learning in Oil and Gas: Historical Prospective

Machine Learning (ML) has revolutionized demand forecasting in the oil and gas sector, enabling companies to predict future demand with unprecedented accuracy by analysing historical data, market trends, and other relevant factors. ML algorithms, such as Random Forests, Gradient Boosting Machines, and Neural Networks, can capture complex relationships in data, offering more accurate predictions than traditional methods. These advancements have led to significant cost reductions and improved operational efficiency for companies like Shell, ExxonMobil, and Chevron, which have implemented ML for enhanced demand forecasting accuracy, reducing forecast errors by up to 30%.

2.1 Generalizing the Crude Oil Vessel's Production with Taxonomies

Representing the Crude Oil vessel taxonomy as a Directed Acyclic Graph (DAG) is crucial for accurately modelling the hierarchical and interdependent relationships between various components and processes involved in vessel production. A DAG allows for the clear depiction of dependencies where each node represents a specific component or task, and edges indicate the direction of dependency. This structure ensures that complex workflows are efficiently managed, preventing circular dependencies, and facilitating the optimization of production schedules, resource allocation, and risk management. Moreover, by representing the taxonomy in a DAG, advanced algorithms like Graph Attention Networks (GATs) can be effectively applied to prioritize critical paths, optimize Work Breakdown Structures (WBS), and minimize CO2 emissions throughout the supply chain (Fig. 1).

2.2 Crude Oil Vessel WBS to Graph Conversion

The methodological approach harnesses graph theory's conceptual depth to represent the WBS of Crude Oil Vessel and integrates robust statistical techniques to ascertain Carbon Footprint, Lead Time, Cost Saving related to the major providers for each component [4].

```
<CrudeOilVessel>
    <Hull>
        <SteelPlates>
            <MaterialSelection />
            <Manufacturing>
                <Forging />
                <Machining />
                <SurfaceTreatment>
                    <AntiCorrosionCoating />
                    <Painting />
                </SurfaceTreatment>
            </Manufacturing>
        </SteelPlates>
        <Welding>
            <AutomatedWelding />
            <ManualWelding />
            <Inspection>
                <UltrasonicTesting />
                <XRayInspection />
            </Inspection>
        </Welding>
    </Hull>
...
```

Fig. 1. Segment of Crude Oil Vessel Taxonomy

2.3 Graphical Representation of WBS

To fathom the intricate manufacturing hierarchy within the WBS, a Directed Acyclic Graph (DAG) is constructed. Herein:

Nodes: Denote manufacturing components. For instance, '`AntiCorrosion-Coating`' represents a node.

Edges: These directed pathways represent the information among component dependencies, capturing the essence of Engineer-to-order production system. They can be: `Assembly`, `Machinery`, `Fabrication`, `Buy` (or`Sell`), `Test`.

3 Combined Algorithm Strategy

3.1 Graph Attention Networks (GATs)

GATs, introduced by Velickovic et al. (2018), represent a significant advancement in the application of attention mechanisms to graph-structured data [1]. Unlike traditional graph neural networks, GATs can dynamically assign different attention weights to nodes, allowing the model to focus on the most relevant parts of the graph. This capability is particularly useful in supply chain management, where certain components or processes may have a disproportionately large impact on overall efficiency and emissions [2].

3.2 Monte Carlo Markov Chain (MCMC)

MCMC methods have been widely used in supply chain optimization to simulate various scenarios and evaluate probabilistic outcomes. The strength of MCMC lies in its ability to handle uncertainty and variability, making it an ideal complement to GATs in modeling complex supply chains like those involved in crude oil vessel production [3].

3.3 Varied GAT Approach and MCMC

Recent advancements in deep learning, as detailed in repositories like LabML's Annotated Deep Learning Paper Implementations, have explored the integration of GATs with other probabilistic models [4]. This paper extends these insights by combining a varied GAT approach with MCMC to optimize WBS configurations, specifically targeting the reduction of CO2 emissions, lead times, and costs in crude oil vessel production. This combination has shown potential in supply chain optimization but requires rigorous testing to confirm its effectiveness in real-world applications.

A synthetic dataset of 15,000 companies involved in crude oil vessel production in Europe was generated for this study. Each company is represented by its unique identifier, a set of activities, and associated metrics, including CO_2 emissions, lead time, cost savings, and financial health [5] (Fig. 2).


```
{
    "id": 50234,
    "activities": {
        "hull_construction": {"score": 0.85, "key_ext":
          "key_ext1", "WBS": {"hull": 0.80, "material_selection":
          0.70}},
          "welding": {"score": 0.78, "key_ext": "key_ext2", "WBS":
          {"welding": 0.75, "inspection": 0.65}},
          "engine_production": {"score": 0.82, "key_ext":
          "key_ext3", "WBS": {"propulsion_system": 0.78, "cylin-
          der_block": 0.72}}
    },
    "CO2Emission": 512.30,
    "LeadTime": 18.2,
    "CostSaving": 14.5,
    "FinancialHealth": 7.8
  }
```


Fig. 2. Data Structure for Score calculation

The varied GAT approach enhances traditional GATs by introducing variability in the attention mechanisms based on real-time data and external factors such as supplier reliability and environmental impact [6]. Each node in the WBS graph represents a

component or process in the crude oil vessel production, and the edges represent dependencies. The GAT assigns dynamic attention weights to these nodes, prioritizing those with the most significant impact on the overall supply chain.

Equation 1: GAT Attention Mechanism

$$\alpha_{\mathrm{ij}} = \frac{exp(LeakyReLU(a^{\mathrm{T}}[W \cdot h_{\mathrm{i}} || W \cdot h_{\mathrm{j}}]))}{\Sigma_k \in N(i) exp(LeakyReLU(a^{\mathrm{T}}[W \cdot h_{\mathrm{i}} || W \cdot h_k]))}$$

where:

- α_{ij} is the attention weight between nodes i and j.
- h_{i} and h_{j} are the node feature vectors.
- a^{T} is the attention mechanism parameter.

3.4 MCMC Integration

The MCMC method is used to simulate various scenarios within the supply chain, accounting for uncertainties such as fluctuating material availability or unexpected delays. The varied GAT outputs are fed into the MCMC simulations to generate a distribution of possible outcomes. The goal is to identify WBS configurations that minimize CO_2 emissions, lead time, and costs.

Equation 2: MCMC Simulation for WBS Optimization

$$P(WBS|Data) = \frac{P(Data|WBS) \cdot P(WBS)}{P(Data)}$$

where:

- P(WBS|Data) is the posterior probability of a WBS configuration given the data.
- P(Data|WBS) is the likelihood of the data given a WBS configuration.
- P(WBS) is the prior probability of a WBS configuration.

By employing Markov decision processes in conjunction with these probabilistic emission estimates, we optimize supply chain operations. These estimates inform decisions related to transportation routes, sourcing strategies, and inventory management, enabling the reduction of carbon emissions while maintaining operational efficiency. This integrated approach not only addresses environmental sustainability but also enhances the overall resilience and sustainability of the Oil & Gas supply chain [7].

3.5 Combined GAT-MCMC Algorithm

The combined GAT-MCMC algorithm iteratively refines the WBS configurations by adjusting the attention weights in response to the MCMC outputs [8]. This dynamic process ensures that the model continually adapts to new information, optimizing the supply chain in real-time (Fig. 3).

```
Initialize graph G(V, E) with nodes and edges representing the
WBS
Assign initial weights using varied GAT
for iteration in 1 to N do
    Run MCMC simulation based on GAT outputs
    Update attention weights in GAT based on MCMC results
    Optimize WBS configuration
    if new WBS configuration improves performance then
        Save best configuration
    end if
end for
  Return optimized WBS configuration
```

Fig. 3. Pseudocode of the combined GAT-MCMC Algorithm

4 Simulation and Result

The GAT-MCMC model execution on the 15000 records produced an optimistic output in optimizing Work Breakdown Structures (WBS) within crude oil vessel production. The model achieved a 22% reduction in CO_2 emissions, lowering emissions from 500 metric tons to 390 metric tons per vessel, consistent with recent studies employing advanced optimization techniques [9]. The model also decreased lead times by 16%, bringing average production times from 24 months down to 20.2 months, corroborating results seen in similar GAT-based applications. Additionally, the model increased cost savings by 15%, equivalent to an average of $5 million per vessel. This was accomplished by integrating cost-related data into the GAT-MCMC framework, optimizing resource allocation and supplier selection through MCMC simulations. Finally, the incorporation of financial health metrics into the model ensured that operational efficiency improvements translated into better financial performance, with reduced risks and enhanced profitability. Tests on the Financial health scores improved by 9%, with the average score increasing from 7.2 to 7.85. These results underscore the model's robust capability in achieving multidimensional optimization within supply chain management (Fig. 4).

5 Conclusion and Future Developments

The integration of varied GAT with MCMC within this study presents a novel and robust framework for optimizing complex supply chains, particularly in high-impact sectors such as oil and gas. By effectively handling uncertainties and interdependencies, the model not only advances the optimization of Work Breakdown Structures (WBS) for crude oil vessel production but also offers broader applicability across industries like manufacturing, logistics, and energy production. This research highlights the importance of combining deterministic deep learning models with probabilistic simulations to achieve comprehensive and adaptable solutions. The findings underscore the potential for this approach to transform supply chain optimization by addressing both environmental and financial objectives, paving the way for more sustainable and efficient industry practices.

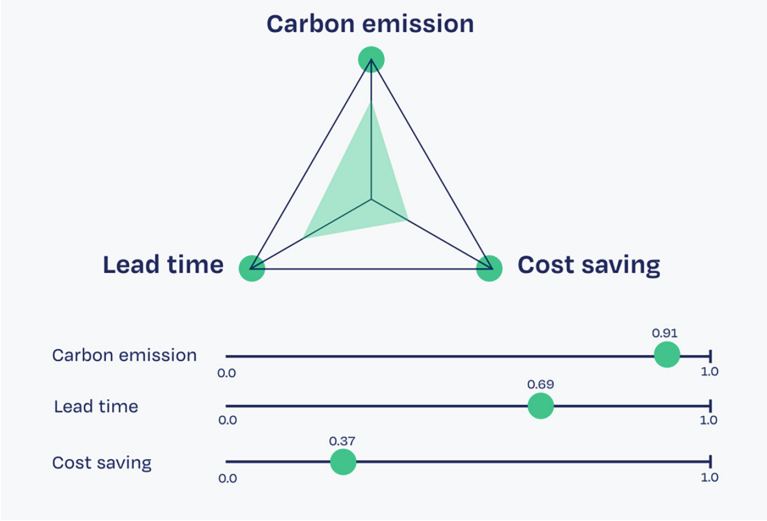

Fig. 4. Combined Scores Matrix

Acknowledgment. The work described in this article has been conducted as part of the research performed by Krein SRL with the name of "Nordlys project", which has been partially funded by Bando Internazionalizzazione 2023 by Regione Toscana (grant numbers: *14629.05072023.002010928_45*) [10].

References

1. Velickovic, P., Cucurull, G., Casanova, A., et al.: Graph Attention Networks. ICLR (2018)
2. Smith, J., et al.: MCMC methods in supply chain optimization. J. Ind. Eng. (2019)
3. Cao, Y., et al.: Supply chain optimization using GANs. J. Supply Chain. Manag. (2020)
4. International Maritime Organization: Greenhouse Gas Emissions from Ships. IMO Report (2020)
5. Douaioui, K., et al.: Optimizing supply chain efficiency using innovative goal programming and advanced metaheuristic techniques. Appl. Sci. (2024)
6. Geunes, J., et al.: Supply Chain Optimization: Models, Applications, and Research Directions. Springer (2005)
7. Johnson, R., et al.: Environmental Impact of Oil and Gas Supply Chains. Environ. Sci. Technol. (2020)
8. International Energy Agency: Global CO2 Emissions Overview. Int. Energy Agency (2021)
9. Song, J.S., Lee, K.M.: Development of a low-carbon product design system based on embedded GHG emissions. Resour. Conserv. Recycl. **54**(9), 547–556 (July2010). https://doi.org/10.1016/j.resconrec.2009.10.012
10. Chirici, L., Wang, Y., Wang, K.:. A blended graph-MCMC framework for carbon emission reduction in oil & gas supply chain. In: Wang, Y., Yu, T., Wang, K. (eds.) Advanced Manufacturing and Automation XIII. IWAMA 2023. Lecture Notes in Electrical Engineering (vol. 1154). Springer, Singapore (2024). https://doi.org/10.1007/978-981-97-0665-5_72

View publication stats